

Table 1. Review of LIANA+ computational methods for cell-cell communication

Method	How it computes interaction	Statistics
cellphonedb	Arithmetic mean of ligand/receptor expression. Multisubunit complexes are taken into account by taking the subunit with lowest expressed	Label permutation
singlecellsignalR	Regularized product score based on squared expression level of ligand and receptor	Based on an estimated threshold (0.5) derived from authors benchmark
cellchat	Law of mass action taking into account ligand, receptors, subunits and cofactors	Label permutation
connectome	Weight_norm is the product of normalized ligand and receptor expression. Weight_scale (specificity) is the average of z-score of ligand expression in sending celltype and z-score of receptor expression in receiver celltype.	-
natmi_specificity	Mean-expression edge weight as product of ligand and receptor expression Specificity-based edge weight as the product of ligand and receptor specificities, defined as their mean expression in the interacting cell type divided by their total sum across all dataset cell types	Ranking of edge weights
geometric_mean	Geometric mean of ligand/receptor expression. Multisubunit complexes are taken into account by taking the subunit with lowest expressed	Label permutation from cellphonedb
scseqcomm	Calculate ligand and receptor scores based on average cluster expression and compare it to a null distribution based on gene permutation within the cluster. The final score for a L-R pair is defined as the minimum of the ligand score in the sender cluster and receptor score in the receiver cluster	Manual score threshold > 0.8 as being significant according the original manuscript
log2fc	Calculates the average one-versus-all log2 fold changes for sender and receiver celltypes.	Manual thresholding > 1 was used